

AccuKnox — IAM Permissions Explorer

Candidate: Sumangal Chhetri

[Figma Prototype — AccuKnox IAM Permissions Explorer](#)

1. Problem Statement & User Persona

Cloud environments often accumulate excessive and unused IAM permissions across users, roles, and workloads. Security engineers need to identify high-risk access, understand whether permissions are actually being used, and reduce unnecessary privileges without disrupting production workloads.

Primary User — Cloud Security Engineer:

The user monitors IAM security across multiple cloud accounts and needs to prioritize risky findings, investigate permission usage, and safely remediate excessive access. Key pain points include large numbers of findings, limited context around permissions, and the risk of breaking workloads when removing access.

2. Proposed Solution

IAM Permissions Explorer follows a simple workflow:

Discover → Investigate → Remediate

- Risk-based Dashboard — Must Have: Identifies high-risk identities and excessive/unused permissions, with filters for account, risk level, finding type, and last-used activity.
- Permission Investigation — Must Have: Compares granted permissions with observed usage so engineers can understand what is actually required.
- Least-Privilege Recommendation — Must Have: Suggests a narrower policy based on observed usage and highlights permissions that can potentially be removed.
- Safe Remediation — Next: Provides before/after policy comparison, impact assessment, approval workflow, and audit trail.

3. Prioritization

The features were prioritized around the core security workflow: Discover → Investigate → Recommend → Remediate → Audit.

The first three capabilities are essential because they allow an engineer to move from identifying a finding to making an evidence-based remediation decision. Safe remediation and audit capabilities follow to make changes safer and more suitable for enterprise environments.

4. Success Metrics

- Reduction in excessive and unused permissions.
- Median time from finding discovery to remediation decision.
- Percentage of recommended remediations accepted.
- Percentage of high-risk findings resolved.
- Percentage of remediation changes completed without workload-related incidents.

5. Bonus — Development Action Items

- Integrate AWS IAM and CloudTrail initially, with future multi-cloud support.
- Build permission analysis and risk-scoring services.
- Generate and validate least-privilege policy recommendations.
- Add policy simulation, approval, rollback, and audit capabilities.
- Test recommendations against representative production-like workloads.